

Attachment Rescue: giving NCI's buried supplementary datasets a permanent home

Submitted by Ochiai, Ojiro · Individual · Narrative, four prompts plus disclosure

PROMPT 1

Significance and Approach

The gap. Before the 2023 NIH Data Management and Sharing Policy, the normal way to share a cancer dataset was to attach it to the paper as "Supplementary Data 3": expression tables, patient cohorts, drug screens, mass spectrometry. Those files still exist in PubMed Central, but NCI's catalogue does not list them and nothing says what is inside. A researcher looking for glioblastoma cohort data with IDH status cannot find a 121-patient hospital cohort attached to a 2018 paper without already knowing the paper; its publisher gave the file a Figshare DOI, yet that record names no NCI grant and NCI's own index has never heard of it.

How big it is. We measured it. Europe PMC lists 58,469 NCI-funded open-access papers from 2016 to 2022; 26,228 have supplementary files and no text-mined reference to any of ten data repositories (GEO, SRA, dbGaP, EGA, ArrayExpress, PRIDE, BioProject, MetaboLights, PDB, GenBank). We drew random samples of 300 papers for each of four cancers (glioblastoma, breast, colorectal, leukemia), downloaded every attachment, and opened every spreadsheet, CSV, Word and PDF file. A file counts as a dataset only if its largest table has at least 50 rows and 3 columns, or 1,000 rows. **The four cancers agree within three points: pooled, 202 of 1,076 papers (19%) hold a data table and 156 (14.5%, confidence interval 12% to 17%) hold it as a native spreadsheet rather than a table inside a PDF.** 167 of the 202 (83%) are CC BY or CC0, so their files can be re-deposited today. Tables ran up to 152,683 rows. The classifier's decisions were re-examined on 90 files (with the AI assistant, not blind; verdicts published): all 60 it called datasets were real data tables, and every miss was under 50 rows. A blind human check is next. We then ran a census of the full population from each paper's full-text attachment list, downloading nothing: 73,795 attachments across 25,138 readable papers. **4,759 papers (18.9%) carry at least one spreadsheet, CSV or tab-delimited text attachment, 15,458 such files in total, 15.9 gigabytes of Excel alone.** 3,824 of those papers (80%) are licensed CC BY or CC0. Then we stopped estimating and opened every one: all 25,411 attachments of the 4,759 spreadsheet papers. **9,543 files in 3,736 papers are datasets by our threshold, 85% of them under a license that permits re-deposit.** About 2,200 (from BMC and PLOS journals) already carry a publisher Figshare DOI;

7,335 have no identifier anywhere. About 15% of population papers do cite a repository our filter missed, checked on 500 papers, so some of these duplicate deposited data. Authors say so themselves: in those 500 papers, 17% of data-availability statements say the data is "in the paper or supplementary files" and 15% say "on request." The papers with buried spreadsheets acknowledge 4,242 NCI grants whose 2016 to 2022 awards total \$11.2 billion (NIH RePORTER); the same pattern across all 24 NIH institutes covers 124,758 papers, on the order of 50,000 buried datasets NIH-wide. And it is not history: in a random 600 NCI open-access papers from 2024 and 2025, under the new policy, 23.2% still carry a spreadsheet attachment with no repository accession, higher than before.

Who is affected. Researchers, because data they need is unfindable; trainees and small labs most, because they lack the network that knows which paper to open. Students, patient advocates and Data Jamboree participants. The patients: verified patient-level tables in our four samples alone describe about 4,400 patient records, roughly 500 from institutional cohorts not in TCGA, GEO or dbGaP. The authors, whose data earns no citations because it has no identifier. And NCI, which paid twice and receives no reuse.

Why it matters to NCI and ODS goals. These papers were compliant. They followed the norms of their time, and the data still vanished from view: the situation the challenge describes when it says policy alone does not drive culture change. The challenge FAQ says a Track 1 "proposal to develop a mechanism for making outputs publicly available is explicitly responsive"; this is that mechanism. Data only insiders can find is not broadly, rapidly or equitably shared. ODS's own workshop report names "incomplete, incorrect, or missing metadata" and the findability of siloed data as major bottlenecks; its Data Jamboree paper names "limited awareness of available datasets" as a primary barrier. A supplementary file is the purest case of both: real data with no metadata, in a silo of one.

The approach. A retrospective rescue over infrastructure that already exists:

1. **Find.** Query Europe PMC for NCI-funded open-access papers with supplementary files and no repository accession; read each paper's full text to list its attachments. One call per paper, no downloads.
2. **Sort.** Download the tabular attachments and classify each: dataset, small table, or not data; primary measurements (cohort tables, expression tables, counts, spectra) versus derived results (gene lists, pathway tables). Skip peer-review files, reporting summaries and figures.
3. **Describe.** Generate a standard dataset record for each, in the DataCite format repositories and search engines read: title, creators and license from the article, article DOI and PMC identifier, funder, format, sheet names, and a one-sentence plain-language description of what the table contains. A person spot-checks.

4. **Deposit or register.** For CC BY and CC0 articles, deposit the unmodified file in one of the free general-purpose repositories NIH already recognizes, which assigns it a permanent identifier (a DOI). For no-derivatives licenses, and for any file whose authors object, register a metadata record pointing to the file's existing location in PubMed Central instead of copying it. Who deposits, and how authors are told, is addressed below.
5. **List.** Deliver the rescued records to the Index of NCI Studies as a curated source file in the format its pipeline already accepts. We have produced that file for all 9,543 datasets, in the index's exact 27-column schema; adding them is one load, not new infrastructure.
6. **Keep alive.** Check every rescued link monthly and publish the result.

Steps 1 through 3 have run on the entire NCI census, and step 4 as far as packaging: eight glioblastoma datasets sit ready for deposit with complete metadata, awaiting the governance decision below. We also put three rescued glioma tables back to work: two TCGA-derived tables from different papers join cleanly on 244 shared patients and, pooled to 1,037, reproduce the known IDH survival split; the third is the hospital cohort above. What remains is deposit, listing and upkeep.

PROMPT 2

Potential Impact on the Cancer Research Community

Immediate. 9,543 existing cancer datasets become findable, citable and described, 7,335 of which have no identifier today. No new data is needed.

Breadth. The gap is not specific to any cancer or to cancer itself. The same filename census on 300-paper samples from five other institutes gives NHLBI 15%, NIDDK 18%, NIMH 19%, NIA 21%, NINDS 24%, against 18% for NCI. Because the pipeline is defined by a funder tag and a date range, any NCI Division, Office or Center, or any NIH institute, can run it unchanged.

For non-data scientists. A rescued dataset arrives with a plain-language description and a stable identifier: a community college instructor can hand students a real patient cohort table with a DOI; a Data Jamboree team can search for "cohort with survival data" and get a hit. We propose offering the collection to the November 2026 NCI Data Jamboree, an ODS event for exactly these users.

For authors and for credit. Each rescued dataset gains an identifier linked to its article, so invisible data can start to accrue citations; repository-linked papers receive up to 25% more citations (Colavizza et al., 2020), a correlation, but one authors get here at no effort.

For policy and infrastructure across NCI. The census gives ODS a per-year, per-journal, per-grant account of how much NCI-funded data was shared only as attachments and how much is recoverable (Nature Communications alone: 889 papers, 3,201 files); our public tool already answers "what did grant P30CA016672 leave behind?" (735 files). By funding mechanism, across the 26,228 census papers, 43% acknowledge an R01-type grant, 47% a P30 Cancer Center grant, 19% a cooperative agreement, 18% a training award and 8% a SPORC: the buried data spans the extramural divisions, the Cancer Centers Program and the training portfolio, and the census can be cut by any of them. That informs guidance, lets program officers report the public benefit of past investments, and gives the Index of NCI Studies its first source of datasets drawn from publications. From 1 October 2026 grantees report data-sharing progress in each progress report (NOT-OD-26-100); a census re-runnable per grant gives program staff an independent check.

PROMPT 3

Innovation and Awareness of Existing Efforts

What exists. - **Large-scale processing of supplementary files has been done.** NLM's FAIR-SMART (Wei, Lu and colleagues, PLOS Biology, 2025) aggregates PubMed Central supplementary files, converts them to a machine-readable form, uses a language model to sort their tables into semantic table types, and serves them through an API for programmatic retrieval; it argues the files should stay in PMC. Gobeill and colleagues (Bioinformatics Advances, 2025) indexed 36.2 million supplementary files for text search; the Europe PMC team (Kafkas et al., 2015) mined supplements for accession numbers. These serve machines and search queries. None decides, by a stated threshold, which files are datasets; none produces a dataset record with identifier, license and provenance; none counts what a funder's grantees left behind; none reaches a person without an API key; none lists anything in NCI's catalogue. FAIR-SMART's standardized tables are the natural upstream for rescue, and its keep-it-in-PMC position is met by our register-don't-copy option. - **Some files already have identifiers, through publishers.** BMC and SpringerOpen deposit each additional file to Figshare with a DOI, and PLOS deposits supporting information per article; those reach the NLM Dataset Catalog and Google Dataset Search and cover about a fifth of our files. Type is set by file extension, not content, and funding follows the publisher's text, so our example cohort is catalogued under another funder with no NCI grant named. The other three-quarters, led by Nature Portfolio at 40% of files, plus eLife, Elsevier and Oncotarget, have no identifier or description anywhere; BioStudies lists files by name and size only. None of these, Figshare records included, appears in NCI's index. We checked our census against Figshare: about a fifth of the files, from BMC and PLOS, already have a Figshare identifier; the rest do not. Even the mirrored files have no

dataset description and do not reach NCI's index, so they need the sorting, describing and listing steps but not the deposit step. - **The scale is documented.** Federer et al. (2018): 69% of PLOS ONE data statements pointed to the paper or supplements, 18% to a repository; Colavizza et al. (2020): repository links in 21% of PLOS and 12% of BMC papers; Federer (2022): DOIs outlast plain links, which is why rescued files get one. - **The catalogue side exists.** The Index of NCI Studies, managed by ODS and built by CBIIT, ingests datasets from GEO, SRA, dbGaP and curated NCI sources through a public pipeline, but nothing from general-purpose repositories or publications. Its team already runs automated enrichment, including a published tool that maps study text to disease codes, so a curated source of publication-derived datasets is a next step, not a new capability.

What is new. The combination, and the direction. We found no effort that has taken a funder's back catalogue, classified which attachments are datasets, described them, given them identifiers, and delivered them into the funder's own catalogue. Extraction and mirroring exist; funder-scoped classification, records and listing do not. Every component is standard; what is new is pointing them backwards at data shared in good faith and lost anyway.

What we are not proposing. Not a new repository, catalogue or metadata standard, and not a replacement for the 2023 policy; rescue is the backfill for everything published before it, and the monitor for what still slips through.

PROMPT 4

Transferability, Sustainability, and Feasibility

Path to adoption. Every component is public and run by someone else: Europe PMC and FAIR-SMART serve the papers and attachments, general-purpose repositories assign identifiers for free, and the Index of NCI Studies pipeline is open source and ingests curated source files. Adoption is one new curated source in that pipeline, a decision inside NCI rather than new infrastructure.

What it would take. Seconds of compute and cents of storage per paper; the census needs no downloads. Human time goes to spot-checking metadata and one conversation with the INS team about the source file, which already exists.

Constraints, stated plainly.

- *Licensing.* 80% of the spreadsheet-bearing papers are CC BY or CC0 and their files can be deposited unchanged. About 9% are CC BY-NC-ND, which forbids modified copies; those get a metadata record pointing at PubMed Central. Author manuscripts outside the open-access subset are excluded. License tags are enforced before anything is copied.

- *Who deposits, and author consent.* No third party silently uploads authors' files. Preferred: NCI or ODS acts as depositor under an institutional repository account, as NIH does for NIH-authored manuscripts in PubMed Central; otherwise authors are notified 60 days before deposit with a one-click opt-out, and objections yield a metadata-only record. Either way the record names the original authors as creators, links the article, carries a route for correction or removal, and NCI never owns the data. Human-participant tables, even de-identified, get a review step before copying.
- *Signal versus noise.* About two-thirds of sampled papers had no tabular data at all; the classifier keeps that noise out, and its accuracy is published. About half of rescued tables are derived results rather than raw measurements, and each record says which.
- *Duplication.* Some attachments duplicate deposited data or already carry a Figshare identifier; records are matched by article DOI and filename first, and an existing identifier is reused and linked to its NCI grant rather than duplicated.
- *The INS intake.* The one dependency on others. If INS declines, the datasets still exist with DOIs in repositories NIH recognizes, and the census still stands.

Why it persists. After the backlog, the pipeline runs quarterly on new papers and monthly on link health, both scheduled checks. What outlives any team is the curated source file and the census, both public and reproducible.

DISCLOSURE

Use of Generative AI

Generative AI (Anthropic's Claude) was used to help write and run the analysis code, to help draft and edit this narrative, and to draft the dataset descriptions. All sampling, classification and counts come from scripts included with the supporting materials and can be rerun from public data; the re-examination of 90 classifier decisions was done with the assistant and is published for audit. All decisions about scope, claims and recommendations are the submitter's own, who takes full responsibility for the accuracy of the content.